MIL-I-23128A(SHIPS)
24 August 1962
SUPERSEDING
MIL-I-23128(SHIPS)
17 January 1962

MILITARY SPECIFICATION

INSULATION BLANKET, THERMAL, REFRACTORY FIBER, FLEXIBLE

1. SCOPE

1.1 Scope.- This specification covers flexible refractory fiber blanket insulation for thermal control at temperatures up to 2000°Fahrenheit (F.).

1.2 Grades.- Insulation blanket shall be of the following grades, as specified (see 6.2):

Grade A - Temperature up to 1200°F.
Grade B - Temperature up to 2000°F.

2. APPLICABLE DOCUMENTS

2.1 Not applicable.

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS
ASTM C-167-50 - Thickness and Density of Blanket or Batt-Type Thermal Insulating Materials, Method of Test for.
ASTM C-177-45 - Thermal Conductivity of Materials by Means of the Guarded Hotplate, Method of Test for.
ASTM D-1448-59 - Micronaire Reading of Cotton Fibers, Method of Test for.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Material.- The insulation shall be composed of refined refractory fibers with or without binders to form a flexible blanket.

3.2 Dimensions and densities.-

3.2.1 Dimensions.- The blanket shall be furnished in the following nominal dimensions, as specified (see 6.2):

	Inches, nominal
Thickness	1/4, 3/8, 1/2, 3/4, 1, 1-1/2, 2
Width	12, 24, 36, 42
Length	36, 48, 96, 288, 300, 600

3.2.1.1 Tolerances.- A minus tolerance of 1/8-inch in width and length, and zero inch for nominal thickness will be permitted. An excess in all dimensions will be permitted.

FSC 5640

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3.2.2 Density.- The weight per square foot at the ordered nominal thickness shall be such as to provide a density of not less than 2.25 nor more than 4.00 pounds per cubic foot for nominal 3.0 pounds per cubic foot blanket, or not less than 5.00 nor more than 7.20 pounds per cubic foot for nominal 6.0 pounds per cubic foot blanket as specified (see 6.2).

3.3 Physical requirements.- The insulation shall conform to the physical requirements shown in table I.

Table I - Physical requirements

Characteristics	Grade A	Grade B
Compressive resistance, to obtain nominal thickness, pounds per square feet (PSF), (maximum)	50	50
Tensile strength, pounds per square inch (PSI) (minimum)	1.0	0.3
Flexibility	No visible rupture	
Fiber diameter, inches $X10^{-4}$ (maximum)	1.5	3.5
Thermal conductivity, B.t.u. - in/ sq/ ft./hr/°F. temperature difference (maximum) at mean temperature of:		
400°F.	0.38	0.50
600°F.	0.55	0.80
900°F.	----	1.05
Changes under soaking heat, 24 hours at indicated temperature °F.	1200	2000
Tensile strength, p.s.i. (minimum)	1.0	0.1
Loss in weight, percent (maximum)	1.0	2.0
Linear shrinkage, percent (maximum)	1.0	2.5
Deterioration of fiber, percent (maximum)	5	30

3.4 Workmanship.- The insulation blanket shall have no visible defects that will adversely affect its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.-

4.2.1 Lot.- For purposes of sampling a lot shall consist of all blanket of the same grade, density, width, length and thickness produced in one plant under essentially the same conditions.

4.2.2 Sampling for visual and dimensional examination.- Sample sheets or rolls shall be selected from each lot of insulation for the examination specified in 4.3.1, with lot acceptance based on table II.

Table II - Sampling for visual and dimensional examination.

Number of sheets or rolls in inspection lot	Sample number of sheets or rolls	Acceptance number (defectives)	Rejection number (defectives)
15 and under	3	0	1
16 to 40	7	0	1
41 to 65	10	0	1
66 to 110	15	1	2
Over 110	25	2	3

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4.2.3 Sampling for tests.- From each lot, 10 pieces 24 by 36 inches shall be selected at random for the test specified in 4.3.2. Each specimen shall be taken from a different sheet or roll.

4.3 Examination and tests.-

4.3.1 Examination.- Each of the sample sheets or rolls selected in accordance with 4.2.2 shall be surface examined, weighed, and measured to determine conformance with the requirements of this specification which do not require tests. Any sheet or roll in the sample containing one or more visual or dimensional defects shall not be offered for delivery, and if the number of defective sheets or rolls in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be cause for rejection.

4.3.2 Tests.- The samples selected in accordance with 4.2.3 shall be subjected to the tests specified in 4.4 to determine conformance with this specification except for thermal conductivity under the conditions specified in 4.3.2.1.

4.3.2.1 Thermal conductivity.- The test of 4.4.7 need only be conducted for one of the following reasons:

(a) If within the three year period preceding the date of invitation for bid the blanket has not been tested and found in compliance with the requirements of 3.3 (table I).
(b) If the blanket offered for delivery is not the same in all respects as that previously tested.

4.3.2.2 Action in case of failure.- If any of the samples tested is found to be not in conformance with this specification, this shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures.-

4.4.1 Conditioning samples.- Specimens for test shall be dried to constant weight by heating from 220° to 230°F.

4.4.2 Compressive resistance.- The resistance required to compress the insulation to nominal thickness shall be determined by either a dead weight or by mechanical loading. The weight shall be added or applied to the center of the loading plate (6 by 6 by 1/8 inch thick and weighing 1-1/4 PSF), and progressively increased until the insulation is compressed to its nominal thickness. The weight added per square foot shall be taken as the compression resistance to obtain nominal thickness.

4.4.3 Density.- The density of this insulation shall be determined in accordance with the method specified in ASTM C167-50, except that the nominal thickness measurement as determined in 4.4.2 shall be used to calculate the density.

4.4.4 Tensile strength.- The tensile strength of the insulation shall be determined with 12-inch wide and 14-inch long specimens. The specimens shall be suspended vertically and held with two clamping devices positioned 12 inches apart. The clamps shall be formed by bolting together at the ends two 14-inch lengths of 1/2 inch pipe. Sponge rubber, 1.4 inch thick, shall be cemented to the outer surface of the pipe in the clamping area to prevent cutting of the specimens. The clamped specimens shall be pulled to rupture attached to the jaws of a tensile testing machine, the lower jaw of which moves at a uniform rate of 1 plus or minus 0.1 inch per minute. The tensile strength shall be taken as the average of three determinations and shall be expressed in PSI of cross sectional area using nominal thickness in the computation.

4.4.5 Flexibility.- Specimens of insulation, 12 by 12 inches by specified thickness, shall be bent over an angle of 90 degrees and examined for rupture on the outside surface. Any minor surface separations shall be disregarded. The radii over which the material shall be bent, shall be as follows:

Grade	Radius, inches
A	2
B	6

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4.4.6 Fineness of fiber.- The diameter of the fiber shall be determined by either of the following methods:

(a) Microscopic.- The diameter of individual fibers shall be determined microscopically on the basis of measuring ten consecutive fibers from each of five clusters selected at random from the insulation. The average of all measurements shall be reported as the fiber diameter.

(b) Air flow.- The air flow method as measured by the micronaire instrument in accordance with ASTM D 1448-59.

4.4.7 Thermal conductivity.- Thermal conductivity shall be determined in accordance with ASTM C177-45.

4.4.8 Deterioration of fibers.- Ten-gram samples composed of 5/8-inch diameter plugs cut from the insulation, as with a cork borer, shall be used in the fiber deterioration test. The samples shall be weighed, placed in a covered 8-inch diameter 25 mesh sieve over a retaining pan secured in a standard laboratory testing sieve on a shaker operated with a horizontal eccentric circular motion at 285 plus or minus 10 revolutions per minute and a tapping action of 150 plus or minus 5 strokes per minute. After shaking 10 minutes with the tapper in operation, the material collected in the retaining pan shall be weighed and the percentage of the total sample calculated. The difference between values obtained with samples of the insulation in the as received condition and those after the soaking heat tests shall be used to express the fiber deterioration of the insulation.

4.4.9 Physical changes under soaking heat.- Specimens shall be weighed and measured. Then the specimens shall be placed in an electrically heated oven and subjected to the maximum temperature for the respective grade for 24 hours. The specimens shall be removed from the oven and tested to determine conformance with the requirements specified in table I.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early material use.-

5.1.1 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier, at lowest rate, and will afford protection against physical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.2 Marking.- Shipment marking information shall be provided on shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.2).

5.2.1.1 Packing.-

5.2.1.1.1 Level A.- Insulation blanket shall be packed in a fiberboard box conforming to class 2 of PPP-B-636, except that limitations on inside dimensions of box shall not apply. All corners and edge seams, and manufacturer's joint shall be waterproofed with tape in accordance with appendix to PPP-B-636.

5.2.1.1.2 Level B.- Insulation blanket shall be packed in fiberboard boxes conforming to class 1 of PPP-B-636, except that limitations on inside dimensions of box shall not apply.

5.2.2 Marking.- In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

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6. NOTES

6.1 Intended use. - Grade B material is intended for use for thermal control of machinery and equipment such as steam or gas turbines, boilers, and piping at temperatures up to 2000°F. Grade A material is intended as a lower-temperature outer layer component for grade B material, or as insulation where service temperatures do not exceed 1200°F.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of specification.
(b) Grade and class required (see 1.2).
(c) Thickness, length and width required (see 3.2).
(d) Nominal density required (see 3.2.2).
(e) Packing or marking requirements other than required by 5.1 (see 5.2).

6.3 Commercial sizes. - Commercial sizes and thicknesses of insulation covered by this specification are industry standards. Other sizes and thicknesses may be obtained upon request from the procuring activity.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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